sesa lab
SOFTWARE ENGINEERING
SALERNO

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https://sesalabunisa.github.io/

Sustainable Software Engineering

GROUP PROJECT



Sustainable Software Engineering - The Project of the Course

Starting Point

- Based on a **previous project** (e.g., from the Software Engineering course).
- The project must be a **web application** that **uses Git** for version control and should have 5/6 contributors.
- Group Project 2/3 people

Goal

Apply the concepts and tools learned in this course to make your project more sustainable — technically, socially, and environmentally — using the tools introduced during the lectures. If your project includes an AI component, evaluate its fairness metrics.

Future Integration

This project can be extended for **SE4AI (Software Engineering for AI)** next semester, where you'll **build a ML pipeline inside the project or re-engineer the one you already have**, focusing on **MLOps**, **based on the project you choose in this course.**

Sustainable Software Engineering: Deliverables

1. **Technical Report (main artifact) - Before the discussion**
   Description of the original project and sustainability goals.
   Tools applied and methodology used.
   Results of energy, performance, and social analyses.
   Discussion of improvements and limitations.
   Optional: Fairness metrics (if AI included).
2. **Practical Evidence (Inside the report)**
   Screenshots, logs, or outputs from tools.
   Graphs or tables of measurements (before/after).
   Code snippets showing refactoring or improvements.
3. **Git Repository (Link)**
   Full commit history.
4. **Reflection / Sustainability Statement (Inside the report - 1 page)**
   What did you learn about sustainable software engineering?
   What trade-offs did you face (e.g., performance vs. maintainability)?
   How does this project connect to SE4AI or long-term software quality?
5. **Presentation - Discussion (15 min (12 min + 3 min Q&A))**
   Focused on process, findings, and key insights.
   Optional live demo of tool usage or performance measurement.

Sustainable Software Engineering: Evaluation

Description of the original project and sustainability goals	Energy and Performance assessment and improvement	Social assessment and Improvement	Legal assessment and Improvement	Reflection / Sustainability Statement
6 points	**12 points**	**6 points**	**3 points**	**3 points**
Clear and comprehensive description of the original software project, including its purpose, main functionalities, architecture, technologies, and relevant characteristics. Identification and motivation of the main sustainability goals to be addressed during the project, explaining what aspects should be improved, why they are relevant, and how they relate to the different sustainability dimensions and the SDG considered in the course.	Comprehensive assessment of the energy consumption and performance of the original project using appropriate tools and metrics. Identification and implementation of meaningful improvements, followed by a comparison of the results before and after the changes.	Assessment of the social sustainability of the project, with a focus on community smells and collaboration and communication anti-patterns within the development team. Identification of relevant socio-technical issues and proposal of appropriate mitigation strategies or improvements to support healthier and more sustainable collaboration.	Assessment of relevant legal aspects of the project, such as software licenses, dependencies, privacy, or compliance issues. Identification of potential risks and proposal or implementation of appropriate improvements.	Critical reflection on the overall sustainability of the project, summarizing the improvements achieved, remaining limitations, possible trade-offs, and opportunities for further improvement.

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